Asymmetric Process-Reward Adaptation for Mathematical Reasoning

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Abstract

Correct answers can conceal faulty derivations. We propose asymmetric process reward model (PRM) adaptation: retain failure supervision and public step labels while omitting positive outcome supervision. Four matched GRPO experiments test whether this improves mathematical reasoning.

1 Motivation and research question

Answer-correct solutions can contain reasoning errors (Zheng et al., 2025), so outcome supervision may teach a PRM to certify flawed derivations. We ask: **can withholding positive outcome supervision improve downstream policy accuracy?** Reduced false certification must outweigh lost useful supervision. If effective, the rule would adapt public PRMs without new annotations; the controls test the cost and benefit of discarding weak positive supervision.

2 Related work and contribution

PRM800K provides costly human step labels on previously sampled solutions (Lightman et al., 2023); their coverage need not track a changing policy. PRIME updates implicit rewards online from outcome labels (Cui et al., 2025), which do not identify valid individual steps. PURE's min-form credit assignment mitigates but does not eliminate PRM reward hacking (Cheng et al., 2025). LCA learns PRMs with softmax-weighted pooling and explicitly tests answer-correct/process-wrong label noise (Jia et al., 2026). Our contribution is **withholding weak positive labels while retaining step supervision**, tested through downstream GRPO. C adopts LCA-inspired pooling, not its full method; D–C isolates the removed loss term.

3 Proposed method

Offline adaptation. Sample four solutions on each of 512 training problems from the initial policy; retain complete, unambiguously verified trajectories. For native step-boundary correctness scores $p_{\phi,t}$, set $e_{\phi,t} = 1 - p_{\phi,t}$. Following LCA,

$$w_t = \frac{\exp(e_{\phi,t}/T)}{\sum_j \exp(e_{\phi,j}/T)}, \qquad E_\phi(\tau) = \sum_t \mathrm{sg}(w_t) e_{\phi,t}, \qquad T = 0.2. \tag{1}$$

Here sg stops gradients through weights. Define $L_{\text{fail}} = \mathbb{E}_{\tau^-}[-\log E_\phi(\tau^-)]$ and $L_{\text{success}} = \mathbb{E}_{\tau^+}[-\log(1 - E_\phi(\tau^+))]$, taking separate means over failures/successes. A shared set of 2,048 public positive/negative step labels supplies mean binary cross-entropy L_{step}:

$$L_c = L_{\text{fail}} + cL_{\text{success}} + L_{\text{step}}, \qquad c_C = 1, \quad \boxed{c_D = 0 \text{ (ours)}}. \tag{2}$$

Failure labels weakly indicate an error without locating it. Step labels discourage blanket rejection but cannot guarantee it is avoided.

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Policy training. Freeze each PRM throughout RL. For complete, parseable responses in B–D, compute

$$R_i = y_i + 0.2 \min_t p_{\phi,t}(\tau_i), \qquad \hat{A}_i = \frac{R_i - \bar{R}}{\sigma_R + 10^{-4}}. \tag{3}$$

Here y_i is binary answer correctness; $\bar{R}, \sigma_R$ are the group mean and sample standard deviation. A uses $R_i = y_i$; invalid or truncated outputs receive zero in all arms. GRPO applies detached sequence advantages to completion tokens (Shao et al., 2024). Correct rollouts retain outcome rewards. In same-outcome valid groups, $\hat{A}_i = (m_i - \bar{m})/(\sigma_m + 5\times10^{-4})$, where $m_i = \min_t p_{\phi,t}$: the coefficient 0.2 does **not** bound PRM influence after normalization.

4 Experimental design and expected findings

Arm	Reward and PRM preparation	Question tested
A	Outcome only; no PRM	Reference GRPO performance
B	Outcome + original frozen PRM	B–A: value of an existing PRM
C	Outcome + PRM adapted with $c = 1$	C–B: complete adaptation package
D (ours)	Outcome + PRM adapted with $c = 0$	**D–C: value of the proposed change**

Models and controls. Use Qwen2.5-Math-1.5B (policy LoRA rank 16) and the H4 1.5B PRM (PRM LoRA rank 8) (Hugging Face H4, 2025). TRL 0.23.1/PEFT uses GRPO with group reward scaling (Hugging Face, 2026); outcome-only execution is verified on one 80 GB A100. Add a **PRM-only anchor control S**, trained with L_{step} under matched initialization, anchor batches and updates: S–B tests anchor adaptation; D–S tests failure supervision. It adds no RL arm.

Data. Use 6,993 MATH training and 500 validation problems (Hendrycks et al., 2021), MATH500 for primary evaluation, and GSM8K for transfer (Cobbe et al., 2021). Exact normalized question overlap with the public PRM800K inventory is zero for both tests. PRM800K includes 4,500 original MATH test questions (OpenAI, 2023); full MATH is supplementary only. Exclude validation/test/diagnostic questions from new anchors; inherited exposure and paraphrases remain limitations.

Budget and matching. C/D share PRM initialization, batches, optimizer and update count; only c changes, with no loss re-averaging. All arms share policy initialization, prompt order and decoding: eight rollouts, temperature 1.0, 2,048-token cap. Target three paired seeds at **24M completion tokens/run** (288M total); retain 6M/12M/24M milestones. Count failed/truncated outputs and report updates, prompts, lengths and total GPU hours including PRM costs.

Evaluation and power. Use identical greedy decoding without reranking for the initial policy and all milestones. Primary: **D–C MATH500 accuracy at 24M**; D–A/B tests practical value. Report paired-seed differences, mean/std, and 95% question-cluster bootstrap intervals retaining all seeds together. At an assumed 10% disagreement rate, the approximate single-seed 80%-power detectable difference is 4 percentage points (two-sided $\alpha = .05$). Small effects may remain unresolved; milestones do not select the endpoint.

Mechanism and risks. On ProcessBench MATH, report first-error accuracy, correct-chain acceptance and their harmonic mean at threshold 0.5, plus valid-chain AUROC among answer-correct solutions (406 valid/94 flawed before overlap stratification). These secondary diagnostics disclose inherited overlap and use no tuning. Log all-correct/all-wrong/mixed group rates and advantage magnitudes, flag invalid outputs, and monitor step counts and missing delimiters under a fixed double-newline splitter. PRM metrics alone cannot establish policy gains; reward hacking and policy drift remain risks.

Tentative timeline. Week 1: buffer, anchors, PRM controls and throughput pilot. Weeks 2–3: prioritize paired C/D runs, then remaining A/B runs within budget. Week 4: diagnostics, statistics and report; disclose unfinished runs.

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